Marina Khismatullina

From:	Journal of Business & Economic Statistics <onbehalfof@manuscriptcentral.com>
Sent:	zaterdag 21 september 2024 20:15
To:	Marina Khismatullina
Subject:	Journal of Business & Economic Statistics - Decision on Manuscript ID JBES-P-2022-0567.R1
Attachments:	Attached standard file: Vogt - JBES.pdf

21-Sep-2024

Dear Dr. Khismatullina:

I have now heard from the Associate Editor regarding your manuscript, "Multiscale comparison of nonparametric trend curves" [Paper JBES-P-2022-0567.R1], that you submitted to the Journal of Business & Economic Statistics.

I apologize for the delay, which is entirely my fault. Referee 1 signed off. However, Referee 2 (whose comments appear at the bottom of this email message) is not satisfied with the revision and doubles down on their previous comments. I believe it is straightforward to implement the suggestions. The AE (whose comments are in the attached PDF file) raises more substantial points.

I'd be happy to consider a revision of your paper, though with no guarantee that it will ultimately be accepted. Please include a response to the AE and reviewers' comments with your revision.

When uploading your response file, please choose the "Author Response to Reviewers" file designation from the drop-down menu. Please be sure to blind your response.

Your paper might be rejected if we do not receive a revised manuscript by 16-Sep-2025. Also, note that before a paper can be published, it will need to conform to the ASA and JBES style and formatting requirements. These requirements include no footnotes, double-spacing your paper, no color figures, and an acknowledgments section.

To submit a revision, please click on the link below:

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IMPORTANT: Your original files are available to you when you upload your revised manuscript. Please delete any redundant files before completing the submission.

Regardless of whether you decide to resubmit, I would like to thank you for sending your work to the Journal of Business & Economic Statistics.

Sincerely,

Atsushi Inoue

Joint Editor, Journal of Business & Economic Statistics
atsushi.inoue@vanderbilt.edu

Reviewer(s)' Comments to Author:

Reviewer: 2

Comments to the Author
Thank you very much for your very careful revision. I have just one remaining "cosmetic" question: can you explain why in the third row of your figures such as Figure 1, page 27, we see several intervals one op top of the other? In other words is there a meaning of the y-axis of this lower plots, what is its scale? Thank you in advance for adding this information (which I might have missed elsewhere?).

Reviewer: 1

Comments to the Author
Review of JBES-P-2022-0567.R1: “Multiscale comparison of nonparametric trend curves”

To address some “big picture” concerns about practical relevance, the authors clarify that rather than the null hypothesis of “no differences ever”, the main focus is on the alternative hypotheses, and specifically the ability to detect 1) which series are different and 2) when. This is a reasonable response. However:

- The simulation studies do not really showcase these priorities. First, simulation results do not appear at all in the main paper. If there is a question about practical relevance of the methods (which the review team did raise), then empirical support should be elevated above some of the technical details. Second, the global size and power tables do not clearly illustrate such local testing capabilities—these metrics would require a much larger collection of data-generating processes to measure performance across a large spectrum of alternative hypotheses. Third, the suggested competitors have not been included, and no compelling reasons were given (more below). Finally, the authors have kept m_i = 0 to simulate data under the null “WLOG”. As I noted before, this is not WLOG for all methods; that it *is* WLOG for their method suggests that they may observe comparative gains over other approaches for smaller sample sizes (n).

- I do not understand why the authors are resistant to add the simple competitors suggested (additive models w/ CI-based testing and Bayesian analogs). Their reasoning was 1) these methods do not offer the same theory guarantees and 2) they include another method, sizer, and a simplified version of their approach. For 1): if the theory actually offers practical gains, then this should show up clearly empirically and further advertise the proposed methods! For 2): sizer is a reasonable competitor (for n=2), but I don’t see why that would eliminate consideration of any alternatives. If the question is “which curves are different and when”, a natural first approach would be to fit a nonparametric model to each curve and see where/when the confidence/credible intervals do not overlap (w/ or w/o a Bonferroni correction). Beating this approach will be more convincing for general readers.

- Stochastic volatility (SV) is widely important in the types of applications that the authors mention. Yet the authors 1) claim that extensions to SV would be challenging, 2) elect not to consider any simulation designs with SV to examine robustness, and 3) claim that SV can be handled with log-transformations (and then using the same mean models as proposed). Perhaps 1) is true, but 2) is important based on the claims in the paper and the relevance for this journal. Finally, 3) is not correct for the most widely-used SV models.

Editor's Comments to Author:

Associate Editor
Comments to the Author:

Very substantial concerns remain, most notably on the part of Referee 2, and myself. The authors should take the next revision very seriously, and refocus the paper on the clustering part, rather than clinging on to their null hypothesis, which I feel is misguided (as I explain in my report).